

ZYRQUEN G11 Core - Threat Vectors & Invariant Protection Audit

Block #849202 | Merkle Root: 909ab814479844d8a14816bed34cdabb07528e18501da86fc4691763a43fa4c68
Seals: 14902 | SSoT Delta 0.00% | 10/10 REAL_HSM | LOCKED_FROZEN_v1.2_LTS

10 Invariant Laws - 10/10 ALL GREEN PASSED:

INV-01 Canonical SSoT Non-Mutation Rule - PASSED - 0 external mutations allowed
INV-02 Merkle Tree Deterministic Binding - PASSED - Verify to Genesis Root 909ab814...
INV-03 Zero Trust Gateway Continuous Auth - PASSED - Dilithium-5 ML-DSA-87 every tx
INV-04 Blast Radius Boundary <2.0% - PASSED - Auto-healing limited
INV-05 Fail-Closed Auto-Defensive Rebuild - PASSED - Tamper -> Quarantine Chamber 02
INV-06 Telemetry Non-Authoritative Isolation - PASSED - Metrics isolated, no write
INV-07 Zero Drift Baseline Consistency - PASSED - SSoT Delta 0.00%
INV-08 14,902 Sealed Blocks Continuity - PASSED - SHA-256 chaining
INV-09 Thai Sovereign Principal Passport Auth - PASSED - EP-SOVEREIGN-01 only
INV-10 12-Stage Forensic Trace Replay - PASSED - 39.4ms < SLA 142ms

Threat Simulation - INV-05 Fail-Closed:

1. Physical HSM Tamper -> Active Zeroization + Fail-Closed -> Chamber 02
2. Bot Signature Forgery -> Sentinel AI Risk 1.00 Block
3. Replay Flood -> Nonce + Merkle inclusion fail -> Drop
4. Hash Distortion -> SHA-256 chain break -> Forensic Ring Flush

Forensic Benchmark: 12-Stage replay 39.4ms | Cryo 14.98 mK | QOps 851.9 | Module 17 V24

Legal Binding: ETA 2544 Sec9 Intent via Merkle Leaf Sig, Sec26 Non-repudiation Dilithium-5 10/10 HSM FIPS140-3 L4,

Evidence-Bound: Integrity != Authenticity != Truth != Legal Admissibility

Sovereign Principal: Yuttaphum Phakphian #EP-SOVEREIGN-01 OMEGA-1